

Same-cohort audit of the NIH patch-selection ablations

Independent technical report based only on the supplied notebooks and run logs. The main comparison uses the final aligned 5,000/1,000/1,000 patient-disjoint cohort; older runs are kept separate.

Best overall: whole-image control

Macro AUROC 0.6698 +/- 0.0078. It exceeds the full selected patch pipeline by 0.0728 AUROC and 0.0391 AUPRC.

Best patch arm: full selected

Macro AUROC 0.5970 +/- 0.0056. It outperforms every aligned patch ablation, but it does not beat whole-image learning.

Most consequential removal

Removing Grad-CAM reduces macro AUROC by 0.0368 and AUPRC by 0.0198 versus full. These are descriptive differences, not paired significance claims.

What can be claimed

Claim status	Audit conclusion
Supported	The report-guided full miner is the strongest aligned patch arm on macro AUROC and macro AUPRC.
Supported	Ranked patches outperform count- and scale-matched random crops, showing that the ranking procedure carries useful signal.
Not supported	The selected-patch pipeline is superior to the matched whole-image classifier. W0 is clearly higher overall and in all five classes.
Not established	That the selected crops are causally sufficient or localized lesions. Weak patch AUROC inherits image labels and is not a localization metric.

Bottom line

The honest conclusion is that patch ranking helps relative to weaker patch-selection variants and to random crops, but the current crop-only representation loses too much global chest context to beat the compute-matched whole-image model. The patch method remains defensible as an interpretability or localization-oriented auxiliary pathway, not as the best classifier in this experiment.

1. Cohort, task, and training volume

Split	Unique images	Patients	Atelectasis +	Cardiomegaly +	Consolidation +	Edema +	Pleural effusion +
Train	5,000	4,127	363	117	105	36	301
Validation	1,000	930	96	20	29	17	105
Test	1,000	288	123	43	81	37	174

Quantity	Count	Interpretation
Unique training images per model	5,000	The same aligned training cohort is reused across every arm and seed.
Model fits	35	Seven arms x five seeds. This is not 175,000 unique training images.
Training epochs	4 per fit	140 model-epochs across the aligned comparison.
Image-bag presentations	700,000	5,000 images x 4 epochs x 5 seeds x 7 arms.
Patch/view tensors	4,200,000	Each image bag contains K=6 crops or whole-image views.

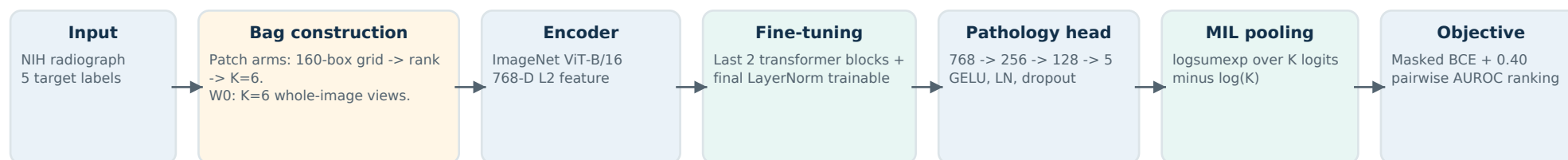
Task definition and fairness controls

- Five independent NIH pathology scores: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. Multi-positive cases are retained.
- Patient-disjoint source splits; the printed integrity checks report 4,127/930/288 train/validation/test patients.
- Unknown labels are maskable in the code. The printed aligned cohort tables show all five labels known for the sampled 5,000/1,000/1,000 rows.
- Model selection is by validation macro AUROC within each seed. The test split is evaluated after loading the best validation checkpoint.
- Macro AUROC is the unweighted mean of five binary one-vs-rest AUROCs. Macro AUPRC is defined analogously.

Evidence for same-cohort alignment

All seven notebooks print the same image/patient counts. A code audit also found identical normalized definitions for split construction, classifier classes, losses, training loop, evaluation, and crossed bootstrap. Only the intended arm configuration and patch-mining path differ.

2. Architecture and operational pipeline



Frozen CheXzero ViT-B/32 is outside this classifier path and is used only to rank patch candidates in patch arms.

Component	Aligned implementation
Classifier backbone	torchvision ViT-B/16 with ImageNet-1K V1 weights; classifier heads removed; normalized 768-D feature.
Trainable backbone	Last two transformer blocks plus encoder final LayerNorm; 14,177,280 trainable encoder parameters.
Pathology head	LayerNorm -> Linear 768-to-256 -> GELU -> Dropout 0.1 -> LayerNorm -> Linear 256-to-128 -> GELU -> LayerNorm -> L2 norm -> Linear 128-to-5; 232,709 parameters.
MIL aggregation	Image logits = logsumexp of the six patch/view logits minus log(6).
Objective	Masked BCEWithLogits + 0.40 pairwise AUROC-ranking loss; gradient norm clipped at 5.0.
Optimizer	AdamW; backbone LR 1e-5, head LR 3e-4, weight decay 0.01.
Transforms	Resize to 224x224, ImageNet normalization; training RandomAffine: +/-3 degrees, 2% translation, scale 0.97-1.03.
Seeds	42, 123, 777, 2027, 31415.

Patch mining in the aligned experiment

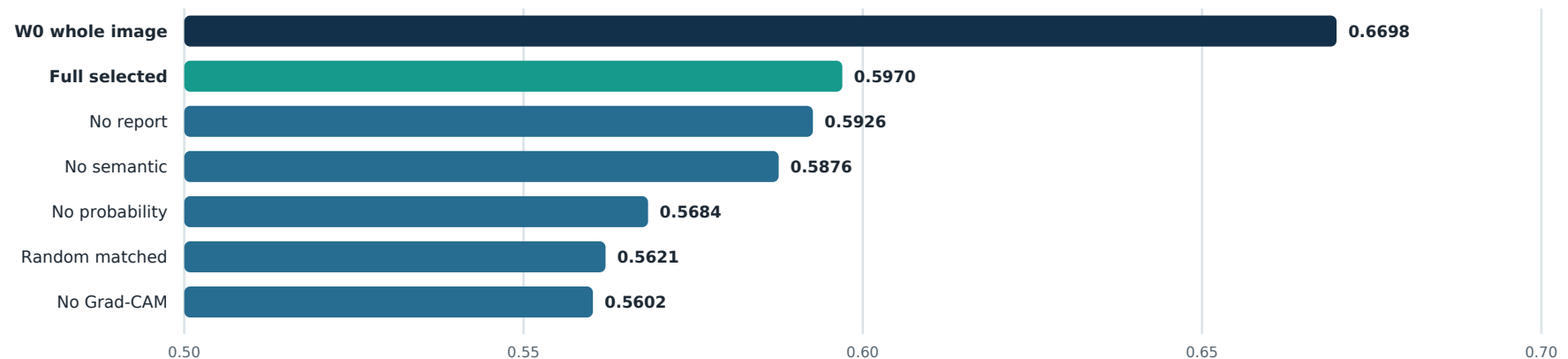
Stage	What the executed code does
Candidate grid	Resize radiograph to 512x512; generate scales 64/128/256 at 50% stride; cap to 160 candidate boxes.
Full training ranking	Frozen CheXzero ViT-B/32 signals combined as a weighted geometric mean: semantic 0.45, probability 0.35, Grad-CAM 0.20. Report phrases guide training mining only when enabled.
Selection	Sort candidate scores; non-maximum suppression at IoU < 0.5; keep K=6 patches.
Validation/test	All patch arms use the same report-free generic image-only miner with semantic + probability + Grad-CAM over the same 160-box grid.
W0	No CheXzero and no crops; six independently augmented whole-image views at train, deterministic whole-image views at evaluation.

Important implementation disclosure

The notebook prints 'Fixed report-free eval boxes/image: 15', because a 15-box helper is constructed. However, the executed `build_common_image_only_eval_patches` function iterates `TRAIN_GRID_BOXES`, which is the capped 160-box grid. This report describes the executed path: 160 candidates, then K=6 selected.

3. Aligned scenarios and overall results

Arm	Only intended change	Macro AUROC	95% crossed-bootstrap CI	Macro AUPRC	Delta AUROC vs full
W0 Whole-image control	No patch mining; six whole-image views, compute matched	0.6698 +/- 0.0078	[0.6352, 0.7052]	0.1737 +/- 0.0078	+0.0728
Full selected pipeline	Report + semantic + probability + Grad-CAM training miner	0.5970 +/- 0.0056	[0.5685, 0.6283]	0.1346 +/- 0.0044	+0.0000
P1 No report	Report guidance removed; generic class prompts retained	0.5926 +/- 0.0058	[0.5644, 0.6255]	0.1318 +/- 0.0032	-0.0043
P2 No semantic score	Semantic similarity removed from patch ranking	0.5876 +/- 0.0096	[0.5575, 0.6178]	0.1322 +/- 0.0040	-0.0094
P4 No probability score	Class probability removed from patch ranking	0.5684 +/- 0.0183	[0.5340, 0.6032]	0.1272 +/- 0.0073	-0.0286
P5 Random matched	Ranked patches replaced by scale- and count-matched random crops	0.5621 +/- 0.0019	[0.5305, 0.5969]	0.1184 +/- 0.0020	-0.0349
P3 No Grad-CAM score	Spatial Grad-CAM score removed from patch ranking	0.5602 +/- 0.0098	[0.5237, 0.5943]	0.1148 +/- 0.0058	-0.0368



Reporting convention: mean +/- sample standard deviation across five independently trained seeds. The 95% intervals are arm-wise crossed training-seed x test-patient bootstrap intervals with 1,200 replicates. They are not paired confidence intervals for differences between arms.

The full selected pipeline is the best patch arm by both macro metrics. W0 is 0.0728 AUROC higher than full selected and its arm-wise bootstrap interval does not overlap the full pipeline interval.

4. Per-class AUROC on the aligned test cohort

Arm	Atelectasis	Cardiomegaly	Consolidation	Edema	Pleural Effusion	Macro
W0 whole image	0.5983 +/- 0.0185	0.6104 +/- 0.0342	0.6833 +/- 0.0202	0.6942 +/- 0.0205	0.7629 +/- 0.0116	0.6698
Full selected	0.5053 +/- 0.0156	0.5432 +/- 0.0191	0.6450 +/- 0.0272	0.6333 +/- 0.0073	0.6581 +/- 0.0100	0.5970
No report	0.5111 +/- 0.0105	0.5455 +/- 0.0100	0.6542 +/- 0.0079	0.6018 +/- 0.0089	0.6507 +/- 0.0103	0.5926
No semantic	0.5099 +/- 0.0108	0.5343 +/- 0.0093	0.6234 +/- 0.0419	0.6163 +/- 0.0184	0.6542 +/- 0.0096	0.5876
No probability	0.5082 +/- 0.0213	0.5261 +/- 0.0133	0.5976 +/- 0.0654	0.5696 +/- 0.0552	0.6402 +/- 0.0088	0.5684
Random matched	0.5026 +/- 0.0075	0.4713 +/- 0.0153	0.6794 +/- 0.0119	0.5960 +/- 0.0178	0.5612 +/- 0.0130	0.5621
No Grad-CAM	0.5027 +/- 0.0150	0.5227 +/- 0.0080	0.6377 +/- 0.0263	0.5836 +/- 0.0082	0.5544 +/- 0.0145	0.5602

Class	W0 whole-image	Full selected	Full minus W0
Atelectasis	0.5983	0.5053	-0.0930
Cardiomegaly	0.6104	0.5432	-0.0672
Consolidation	0.6833	0.6450	-0.0383
Edema	0.6942	0.6333	-0.0609
Pleural Effusion	0.7629	0.6581	-0.1048

Interpretation

- W0 has the highest AUROC for every pathology. The patch deficit is therefore not confined to one rare class.
- The full selected patch arm is strongest among patch methods on Atelectasis and Edema, while some individual ablations are higher on isolated classes (for example, random matched on Consolidation).
- A single class advantage should not be treated as a global win; the declared primary outcome is macro AUROC across all five classes.

5. Per-class AUPRC and secondary weak-patch diagnostic

Arm	Atelectasis	Cardiomegaly	Consolidation	Edema	Pleural Effusion	Macro
W0 whole image	0.1645 +/- 0.0190	0.0800 +/- 0.0246	0.1631 +/- 0.0267	0.0777 +/- 0.0087	0.3834 +/- 0.0319	0.1737
Full selected	0.1404 +/- 0.0056	0.0542 +/- 0.0033	0.1335 +/- 0.0142	0.0663 +/- 0.0119	0.2787 +/- 0.0197	0.1346
No report	0.1281 +/- 0.0033	0.0549 +/- 0.0050	0.1282 +/- 0.0045	0.0529 +/- 0.0026	0.2952 +/- 0.0128	0.1318
No semantic	0.1370 +/- 0.0034	0.0527 +/- 0.0035	0.1259 +/- 0.0103	0.0648 +/- 0.0110	0.2804 +/- 0.0184	0.1322
No probability	0.1360 +/- 0.0053	0.0528 +/- 0.0016	0.1200 +/- 0.0149	0.0623 +/- 0.0170	0.2648 +/- 0.0103	0.1272
Random matched	0.1260 +/- 0.0041	0.0438 +/- 0.0016	0.1495 +/- 0.0060	0.0572 +/- 0.0051	0.2156 +/- 0.0124	0.1184
No Grad-CAM	0.1225 +/- 0.0050	0.0665 +/- 0.0112	0.1238 +/- 0.0105	0.0539 +/- 0.0031	0.2074 +/- 0.0057	0.1148

Arm	Weak patch macro AUROC	Interpretation
Full selected pipeline	0.5647	Patch crop scored against inherited image label
P1 No report	0.5553	Patch crop scored against inherited image label
P2 No semantic score	0.5578	Patch crop scored against inherited image label
P4 No probability score	0.5389	Patch crop scored against inherited image label
P5 Random matched	0.5386	Patch crop scored against inherited image label
P3 No Grad-CAM score	0.5409	Patch crop scored against inherited image label
W0 whole-image control	Not applicable	No patch crops

Do not call the weak-patch score localization accuracy

The weak patch metric assigns each selected crop the image-level label. A high value can arise without spatial overlap with a lesion. The aligned notebooks do not report ground-truth box localization for these 1,000-image test runs, so causal localization is unproven here.

AUPRC is prevalence-sensitive, so it is especially important for the rare Edema and Cardiomegaly classes. W0 remains highest in all five classes. Full selected improves over random matched by 0.0162 macro AUPRC, but remains 0.0391 below W0.

6. What the ablations say about the patch miner

Comparison to full selected	Delta macro AUROC	Delta macro AUPRC	Reading
P1 No report	-0.0043	-0.0028	Report guidance removed; generic class prompts retained
P2 No semantic score	-0.0094	-0.0025	Semantic similarity removed from patch ranking
P4 No probability score	-0.0286	-0.0074	Class probability removed from patch ranking
P3 No Grad-CAM score	-0.0368	-0.0198	Spatial Grad-CAM score removed from patch ranking
P5 Random matched	-0.0349	-0.0162	Ranked patches replaced by scale- and count-matched random crops

Transparent reading

- Grad-CAM has the largest observed component contribution. Its removal is associated with the largest macro AUROC and AUPRC decrease among the named score removals.
- Probability is the next most consequential score in this run. No-probability also has the largest between-seed AUROC variability among the aligned arms.
- Report guidance and semantic similarity have smaller macro effects. Because the report and semantic arms alter related language signals, these are not independent additive contributions.
- Random matching performs below the full miner, supporting non-random ranking quality. However, the random arm exceeds some selected arms on Consolidation, which argues against treating every chosen crop as causally correct.
- No paired difference bootstrap was supplied for these seven final arms. The deltas should therefore be described as observed effect sizes; only W0 versus full has clearly separated arm-wise bootstrap intervals.

Why the whole-image architecture wins

The classifier architecture is not different: normalized function hashes for the ViT encoder, pathology head, MIL aggregation, losses, training loop, evaluation, and bootstrap are identical across all seven notebooks. The decisive difference is the input representation. W0 retains the entire thorax, bilateral symmetry, heart-to-chest ratio, diffuse opacity patterns, and global positional context. The patch arms compress each image into six crops chosen by a frozen zero-shot miner; any missed or ambiguous region becomes unrecoverable to the downstream classifier.

This is especially damaging for global or distributed findings such as cardiomegaly, edema, and pleural effusion, but the per-class table shows a deficit for all five findings.

7. Supplied runs excluded from the same-cohort comparison

The following outputs were supplied and are relevant context, but they cannot be pooled into the 5,000/1,000/1,000 table because their cohort, split, model, metric rows, or task differs. They are reported here to avoid hiding favorable results.

Run family	Reported result	Why it is not in the main table
21 July single-label CheXzero patch discovery	Whole-image CheXzero test: 0.664 AUROC / 0.347 AUPRC; all-images zero-padded patch score: 0.610 / 0.294; queried-only patch score: 0.655 / 0.435; VAL ablation FULL seed 42, n=400: 0.6054 / 0.2882.	8,526/1,564/1,232 train/val/test cohort; zero-shot CheXzero scoring; different rows and queried-only/zero-padding rules; the ablation is one validation subset and one seed.
August RadJEPa/BioViL-T/DDE/RAG pipeline	NIH test macro AUROC: full 0.8908; no report 0.5639; no spatial 0.8916; no fusion 0.6162; no refine 0.8899; unified RAG 0.8892; no spurious 0.8904; ImageNet ViT 0.7729.	Different architecture and 169,051-patch Stage-3 artifact; image labels are attached partly from reports or Stage-3 pathology tags; output itself warns of causal-space collapse and high site leakage. No five-seed uncertainty is shown.
19 August QuantFSL-CSPQ	NIH-to-CheXpert few-shot AUROC about 0.8895 (1-shot) and 0.8882 (5-shot) for the reported FQ-1 setting.	Cross-dataset few-shot task, support-set adaptation, quantum/fusion stack, different target cohort and metric protocol. It also contains an earlier runtime failure before a continuation completed.

Legacy per-class examples (not comparable to the aligned test table)

Legacy setting	Atelectasis	Cardiomegaly	Consolidation	Edema	Pleural effusion
21 July whole-image CheXzero test AUROC	0.605	0.588	0.656	0.764	0.708
21 July VAL FULL, seed 42, n=400 AUROC	0.491	0.700	0.479	0.714	0.643

Source audit

- Direct notebook outputs: fullselectedpipeline.ipynb; p1-noreport.ipynb; p2-nosemanticscore.ipynb; p3-nogradcamscore.ipynb; p5-randommatched.ipynb; w0-wholeimagebudgetmatched.ipynb.
- P4 result source: Pasted markdown (2)(8).md. The provided p4noproabilityscore.ipynb contains code but no embedded outputs.
- Cross-check logs: Pasted markdown(20260825-093827).md (full selected), Pasted markdown (2)(9).md (no report), Pasted markdown(20260824-141924).md (W0).
- Legacy-only sources: nih-1ampatch-21stjuly-smok-5pm (1).ipynb; full-pipeline-patches-train3-auguest.ipynb; pipeline-4gpu-19aug(7).ipynb; Pasted markdown(20260824-144013).md.

8. Final conclusion and recommended next experiment

Evidence-based conclusion

On the final aligned patient-disjoint cohort, the full selected patch pipeline is the strongest patch-only arm (macro AUROC 0.5970 +/- 0.0056), but the compute-matched whole-image control is substantially better (0.6698 +/- 0.0078). Grad-CAM and probability contribute the most within the patch miner. The evidence supports useful patch ranking, not superior classification or verified causal localization.

Recommended next experiment

Priority	Action	Why it is honest and informative
1	Train a dual-branch model: whole-image ViT feature + six selected-patch MIL feature, fused only after both branches produce class logits.	Tests whether patches add complementary local evidence without discarding global context.
2	Add a compute-matched random-patch branch and a whole-image-only branch to the same training script and seeds.	Prevents architecture, optimization, or data-loader differences from explaining gains.
3	Use validation-only selection for fusion weight and patch hyperparameters; lock them before test evaluation.	Avoids tuning on test AUROC.
4	Report paired patient-level bootstrap deltas: fused minus W0, patch-only minus W0, and full-miner minus random matched.	Directly answers whether any gain is statistically credible.
5	Evaluate localization on NIH bounding-box images using pointing game and IoU, with confidence intervals and enough cases per class.	Separates classification utility from spatial/causal claims.

Suggested wording for a manuscript or meeting

"In a controlled five-seed ablation on 5,000 training, 1,000 validation, and 1,000 held-out NIH images, the report-guided semantic/probability/Grad-CAM miner was the strongest patch-only configuration. However, a compute-matched whole-image ViT-B/16 control achieved higher macro AUROC (0.6698 versus 0.5970). We therefore interpret patch selection as a source of local auxiliary evidence rather than a replacement for global image context, and we do not claim causal localization without box-based evaluation."

Audit note: No result was recomputed or altered for this report. Values are transcribed from the supplied embedded notebook outputs and pasted run logs; the code comparison was performed on the supplied source cells.